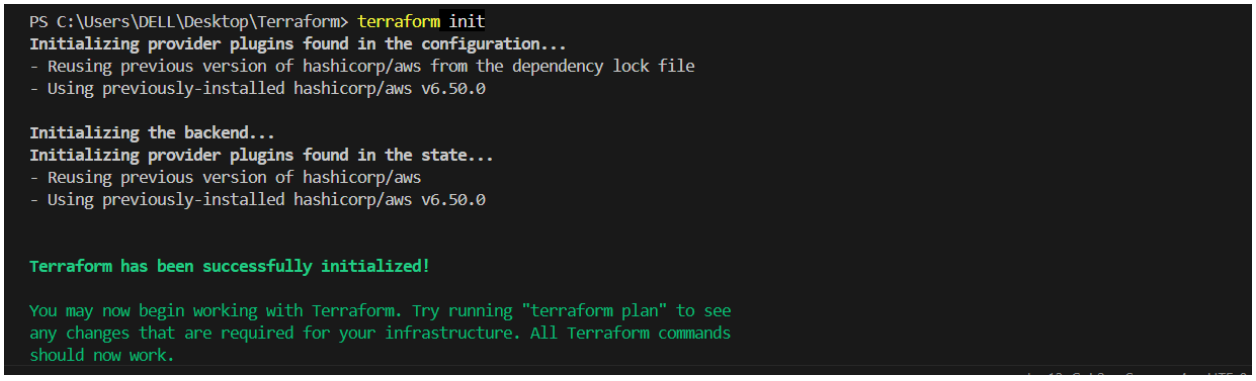
Terraform-05

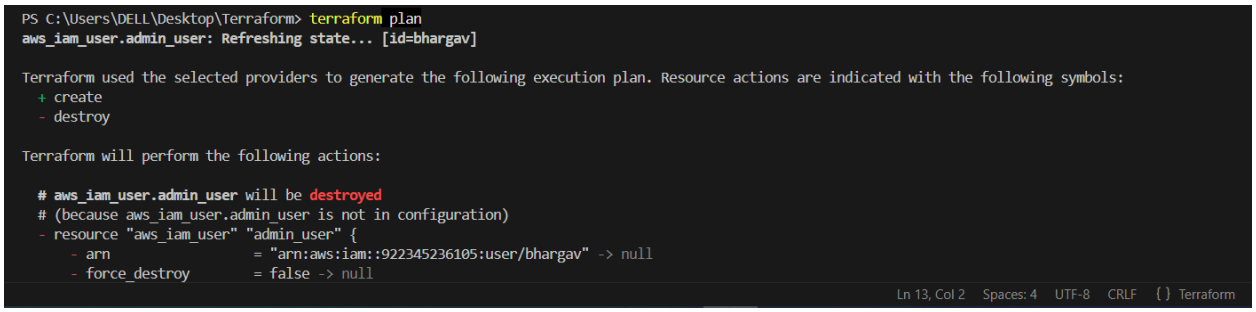
1). Watch the **Terraform-05** video.
2). Execute the script shown in the video.

Write script for create an ec2 instance:

Terraform init:



Terraform plan:



terraform apply:

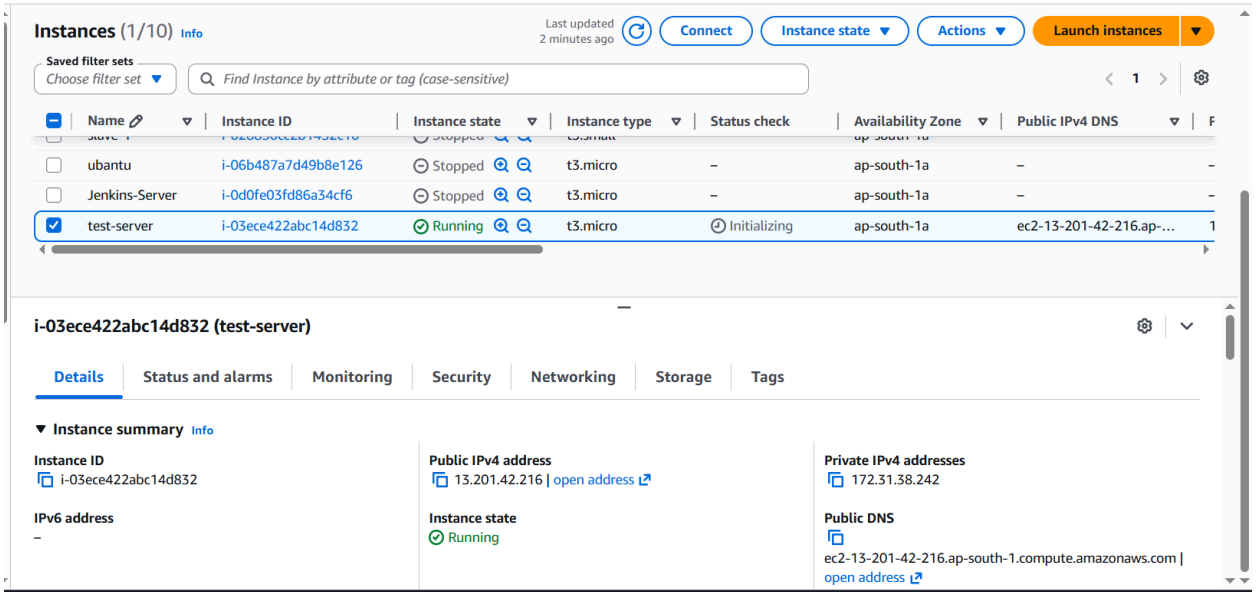
```
PS C:\Users\DELL\Desktop\Terraform> terraform apply
aws_iam_user.admin_user: Refreshing state... [id=bhargav]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create
  - destroy

Terraform will perform the following actions:

  # aws_iam_user.admin_user will be destroyed
  # (because aws_iam_user.admin_user is not in configuration)
  - resource "aws_iam_user" "admin_user" {
      - arn                  = "arn:aws:iam::922345236105:user/bhargav" -> null
      - force_destroy        = false -> null
      - id                   = "bhargav" -> null
      - name                 = "bhargav" -> null
```

Ln 13, Col 2 Spaces: 4 UTF-8 CRLF { } Terraf

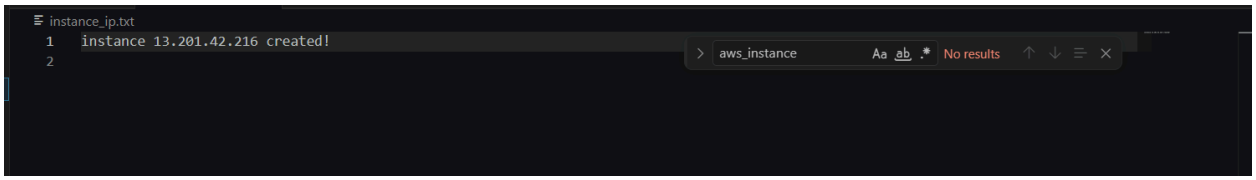
```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.test-server: Creating...
aws_iam_user.admin_user: Destroying... [id=bhargav]
aws_iam_user.admin_user: Destruction complete after 2s
aws_instance.test-server: Still creating... [00m10s elapsed]
aws_instance.test-server: Provisioning with 'local-exec'...
aws_instance.test-server (local-exec): Executing: ["cmd" "/C" "echo instance 13.201.42.216 created! >> instance_ip.txt"]
aws_instance.test-server: Creation complete after 16s [id=i-03ece422abc14d832]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
PS C:\Users\DELL\Desktop\Terraform>
```

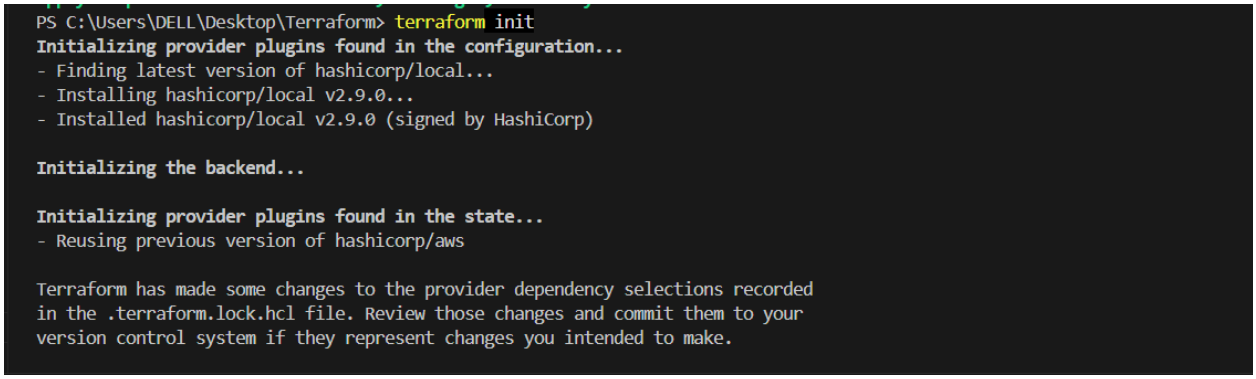
Ln 13, Col 2 Spaces: 4 UTF-8

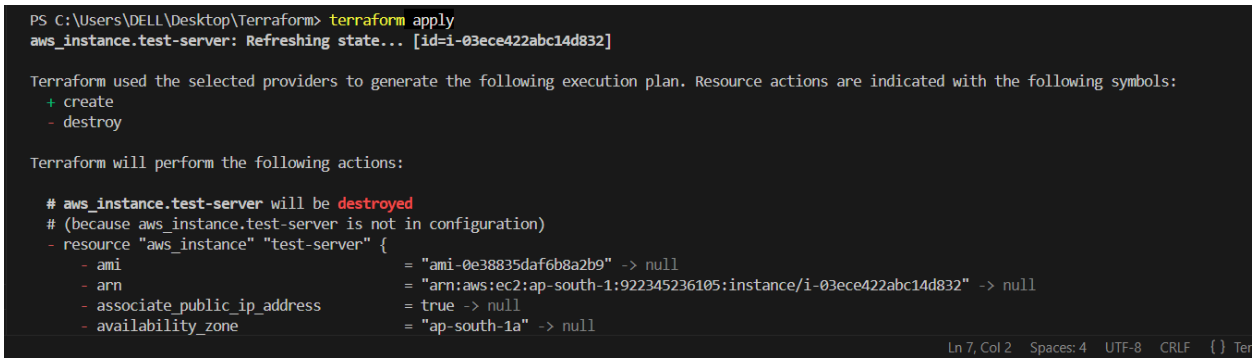
verify:



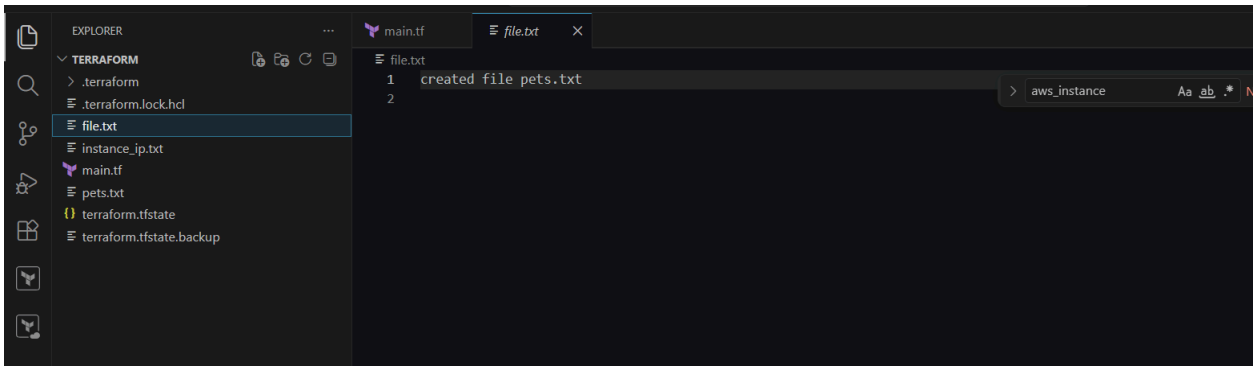
Execute command:







```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                                   powershell

  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_sensitive_file.test: Creating...
local_sensitive_file.test: Provisioning with 'local-exec'...
local_sensitive_file.test (local-exec): Executing: ["cmd" "/C" "echo created file pets.txt > file.txt"]
local_sensitive_file.test: Creation complete after 0s [id=c524a39c02f142ba0b81da289f2e11332d59b4dd]
aws_instance.test-server: Destroying... [id=i-03ece422abc14d832]
aws_instance.test-server: Still destroying... [id=i-03ece422abc14d832, 00m10s elapsed]
aws_instance.test-server: Still destroying... [id=i-03ece422abc14d832, 00m20s elapsed]
aws_instance.test-server: Still destroying... [id=i-03ece422abc14d832, 00m30s elapsed]
aws_instance.test-server: Destruction complete after 30s

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
PS C:\Users\DELL\Desktop\Terraform>
```



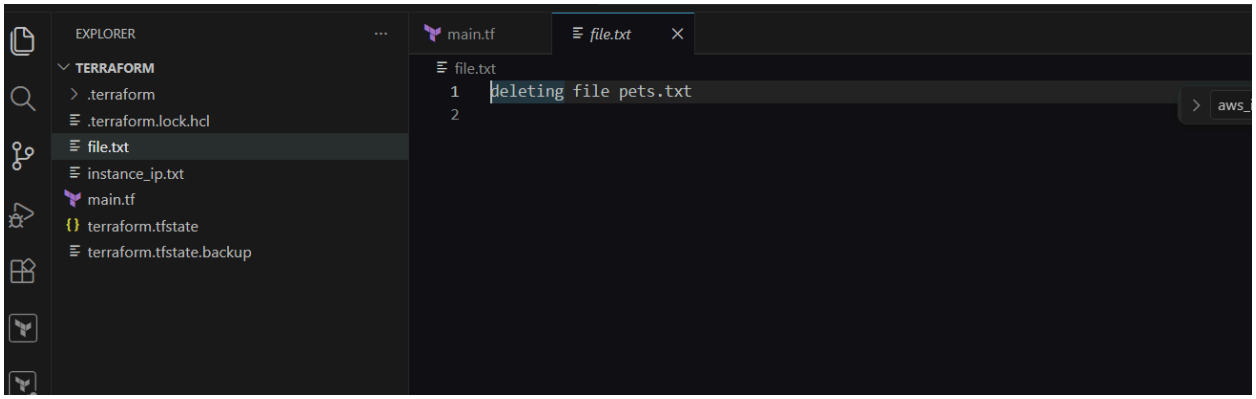
main.tf > resource "local_file" "test" > provisioner "local-exec"

aws_instance

```hcl
resource "local_file" "test" {
  filename = "pets.txt"
  content = "i love dogs"
  provisioner "local-exec" {
    when = "destroy"
    command = "echo destroying file pets.txt > file.txt"
  }
}
```

```
PS C:\Users\DELL\Desktop\Terraform> terraform destroy
local_file.test: Refreshing state... [id=c524a39c02f142ba0b81da289f2e11332d59b4dd]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # local_file.test will be destroyed
  - resource "local_file" "test" {
      - content              = "i love dogs" -> null
      - content_base64sha256 = "Z4bFH8+z8MOLjSNceqSTVrrGES69Br2ia8WdkH64VA8=" -> null
      - content_base64sha512 = "ZS/x0zcNTkZY7S/4HFjwHU12yv1GBtOMocHV5A9+OVHb/Ps8700fdVBd7pAlJtP/no5E2ICD4H4YWHXHOztTMg==" -> null
      - content_md5          = "ef909a06c8df361c85a98135dc8cf42d" -> null
      - content_sha1         = "c524a39c02f142ba0b81da289f2e11332d59b4dd" -> null
      - content_sha256       = "6786c51fcfb3f0c38b8d235c7aa49356bac6112ebd06bda26bc59d907eb8540f" -> null
```

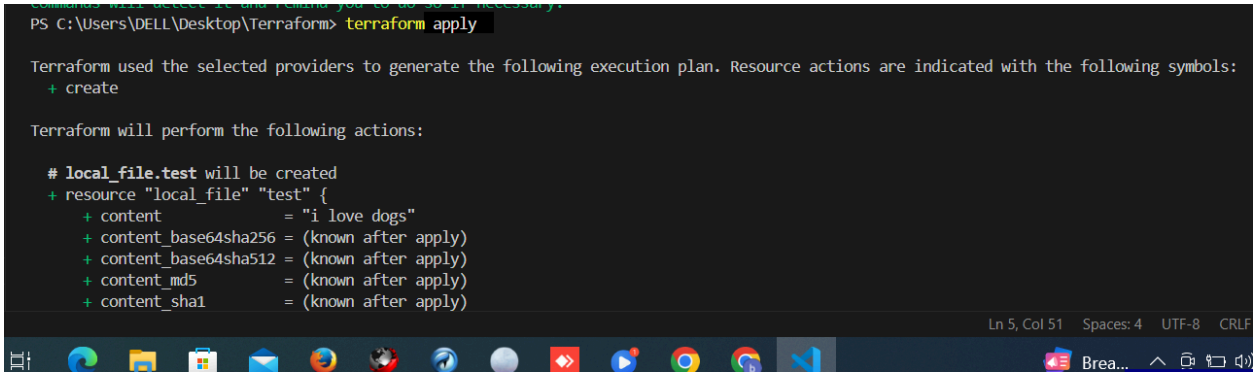
```
Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value: yes

local_file.test: Destroying... [id=c524a39c02f142ba0b81da289f2e11332d59b4dd]
local_file.test: Provisioning with 'local-exec'...
local_file.test (local-exec): Executing: ["cmd" "/C" "echo deleting file pets.txt > file.txt"]
local_file.test: Destruction complete after 0s

Destroy complete! Resources: 1 destroyed.
PS C:\Users\DELL\Desktop\Terraform>
```



Gave a wrong path

Terraform apply:

If we give any wrong path and do terraform apply then in the state file you can


PS C:\Users\DELL\Desktop\Terraform> terraform apply
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
Terraform will perform the following actions:
# local_file.test will be created
+ resource "local_file" "test" {
+ content = "i love dogs"
+ content_base64sha256 = (known after apply)
+ content_base64sha512 = (known after apply)
+ content_md5 = (known after apply)
+ content_sha1 = (known after apply)
Ln 5, Col 51 Spaces: 4 UTF-8 CRLF

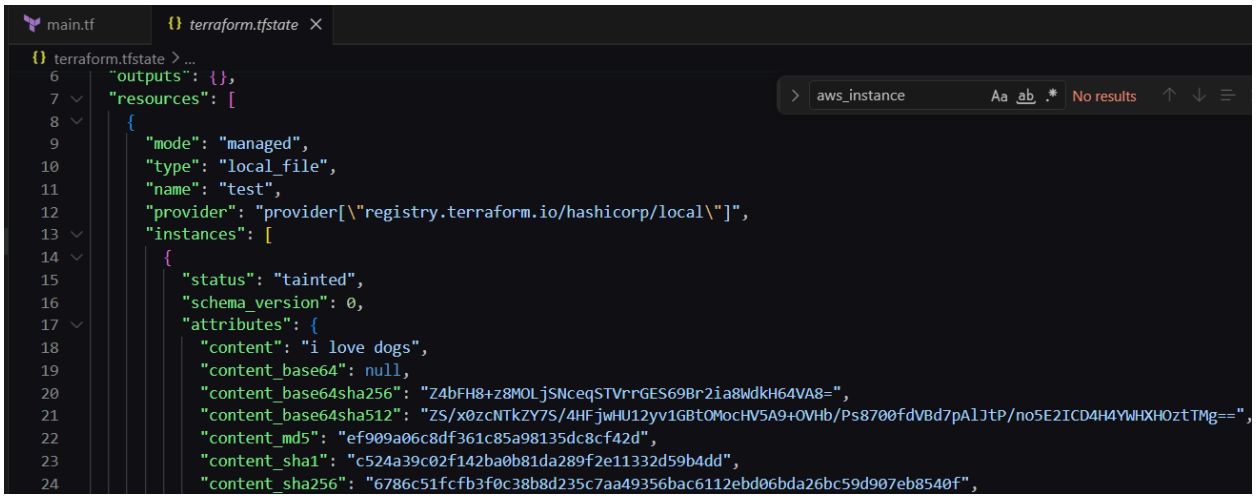

local_file.test: Creating...
local_file.test: Provisioning with 'local-exec'...
local_file.test (local-exec): Executing: ["cmd" "/C" "echo creating file pets.txt > /tmp/file.txt"]
local_file.test (local-exec): The system cannot find the path specified.
Error: local-exec provisioner error
with local_file.test,
on main.tf line 4, in resource "local_file" "test":
4: provisioner "local-exec" {
Error running command 'echo creating file pets.txt > /tmp/file.txt': exit status 1. Output: The system cannot find the path specified.


main.tf
terraform.tfstate
main.tf > resource "local_file" "test" > provisioner "local-exec" > command
aws_instance
No results
resource "local_file" "test" {
filename = "pets.txt"
content = "i love dogs"
provisioner "local-exec" {
command = "echo creating file pets.txt > file.txt"
}
}

tainted:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\DELL\Desktop\Terraform> terraform taint local

Error: Invalid address

  on  line 1:
  (source code not available)

The keyword "local" is reserved and cannot be used to target a resource address. If you are targeting a resource type t
prefix your address with "resource.".

PS C:\Users\DELL\Desktop\Terraform> terraform taint local_file.test
Resource instance local_file.test has been marked as tainted.
PS C:\Users\DELL\Desktop\Terraform>
                                                                                    Ln 5, Col 46  S
```

Status shows tainted



Terraform taint local_file.test
And terraform apply : replaced

```
PS C:\Users\DELL\Desktop\Terraform> terraform taint local_file.test
Resource instance local_file.test has been marked as tainted.
PS C:\Users\DELL\Desktop\Terraform> terraform apply
local_file.test: Refreshing state... [id=c524a39c02f142ba0b81da289f2e11332d59b4dd]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
-/+ destroy and then create replacement

Terraform will perform the following actions:

  # local_file.test is tainted, so must be replaced
-/+ resource "local_file" "test" {
      ~ content_base64sha256 = "Z4bFH8+z8MOLjSNceqSTVrrGES69Br2ia8WdkH64VA8=" -> (known after apply)
      ~ content_base64sha512 = "ZS/x0zcNTkZY7S/4HFjwHU12yv1GBtOMocHV5A9+OVHb/Ps8700fdVBd7pAlJtP/no5E2ICD4H4YWHXHOztTMg==" -> (known after apply)
      ~ content_md5          = "ef909a06c8df361c85a98135dc8cf42d" -> (known after apply)
      ~ content_sha1         = "c524a39c02f142ba0b81da289f2e11332d59b4dd" -> (known after apply)
```

Terraform untaint local_file.test
Terraform apply

```
PS C:\Users\DELL\Desktop\Terraform> terraform taint local_file.test
Resource instance local_file.test has been marked as tainted.
PS C:\Users\DELL\Desktop\Terraform> terraform untaint local_file.test
Resource instance local_file.test has been successfully untainted.
PS C:\Users\DELL\Desktop\Terraform> terraform apply
local_file.test: Refreshing state... [id=c524a39c02f142ba0b81da289f2e11332d59b4dd]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
PS C:\Users\DELL\Desktop\Terraform>
```

ERROR:

main.tf > resource "local_file" "test" > provisioner "local-exec" > command

```
1  resource "local_file" "test" {
2    filename = "pets.txt"
3    content = "i love dogs"
4    provisioner "local-exec" {
5      command = "echo creating file pets.txt > file.txt"
6    }
7  }
```

set-Item -path env:TF_LOG -value “ERROR”

```
PS C:\Users\DELL\Desktop\Terraform> set-Item -path env:TF_LOG -value "ERROR"
PS C:\Users\DELL\Desktop\Terraform> terraform plan
local_file.test: Refreshing state... [id=c524a39c02f142ba0b81da289f2e11332d59b4dd]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
PS C:\Users\DELL\Desktop\Terraform>
```

Import:

main.tf | terraform.tfstate

main.tf > resource "aws_instance" "manual" > ami

```
1  resource "aws_instance" "manual" {
2      ami ="ami-0f5ee92e2d63afc18"
3      instance_type = "t3.micro"
4
5  }
```

Terraform init:

```
PS C:\Users\DELL\Desktop\Terraform> terraform init
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.50.0...
- Installed hashicorp/aws v6.50.0 (signed by HashiCorp)

Initializing the backend...

Initializing provider plugins found in the state...
- Reusing previous version of hashicorp/local

Terraform has made some changes to the provider dependency selections recorded
in the .terraform.lock.hcl file. Review those changes and commit them to your
version control system if they represent changes you intended to make.
```

Terraform import aws_instance.manual <instance id>

```
PS C:\Users\DELL\Desktop\Terraform> terraform import aws_instance.manual i-0d0fe03fd86a34cf6
aws_instance.manual: Importing from ID "i-0d0fe03fd86a34cf6"...
aws_instance.manual: Import prepared!
  Prepared aws_instance for import
aws_instance.manual: Refreshing state... [id=i-0d0fe03fd86a34cf6]

Import successful!

The resources that were imported are shown above. These resources are now in
your Terraform state and will henceforth be managed by Terraform.
```

Terraform destory:

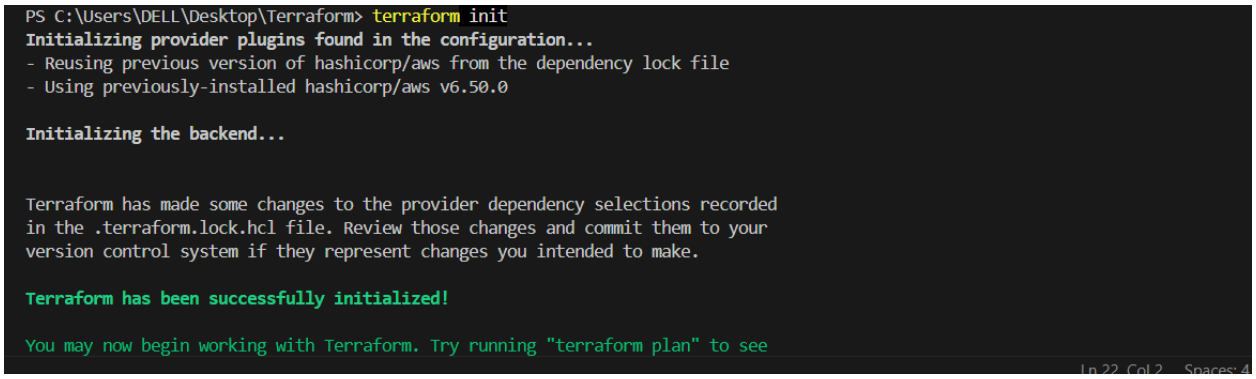
```
PS C:\Users\DELL\Desktop\Terraform> terraform destroy
local_file.test: Refreshing state... [id=c524a39c02f142ba0b81da289f2e11332d59b4dd]
aws_instance.manual: Refreshing state... [id=i-0d0fe03fd86a34cf6]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # aws_instance.manual will be destroyed
  - resource "aws_instance" "manual" {
      - ami                                  = "ami-0f5ee92e2d63afc18" -> null
      - arn                                  = "arn:aws:ec2:ap-south-1:922345236105:instance/i-0d0fe03fd86a34cf6" -> null
      - associate_public_ip_address          = false -> null
      - availability_zone                    = "ap-south-1a" -> null
      - disable_api_stop                     = false -> null
```

```
Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value: yes

local_file.test: Destroying... [id=c524a39c02f142ba0b81da289f2e11332d59b4dd]
local_file.test: Destruction complete after 0s
aws_instance.manual: Destroying... [id=i-0d0fe03fd86a34cf6]
aws_instance.manual: Still destroying... [id=i-0d0fe03fd86a34cf6, 00m10s elapsed]
aws_instance.manual: Destruction complete after 11s

Destroy complete! Resources: 2 destroyed.
PS C:\Users\DELL\Desktop\Terraform> ^C
PS C:\Users\DELL\Desktop\Terraform>
```

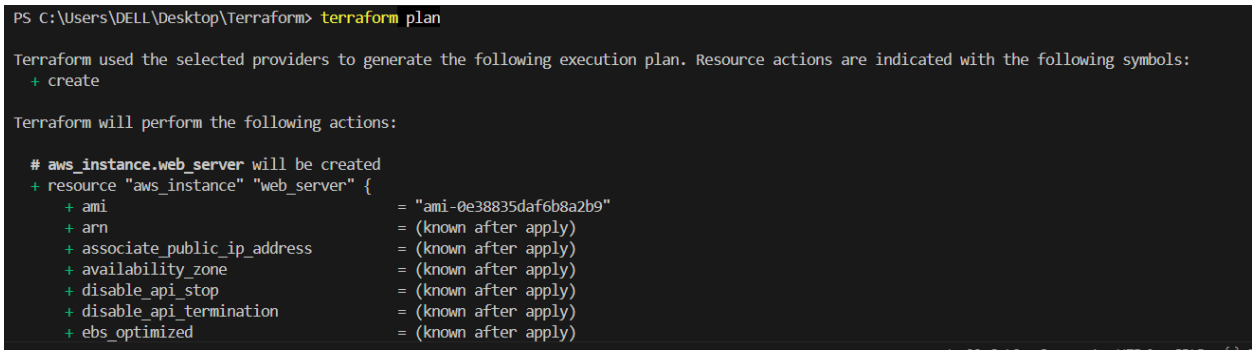
3). Create one **EC2 instance** with **httpd installed** using a Terraform script.

Write a script for create ec2 instance with httpd:

Terraform init:



Terraform plan:



Terraform apply:

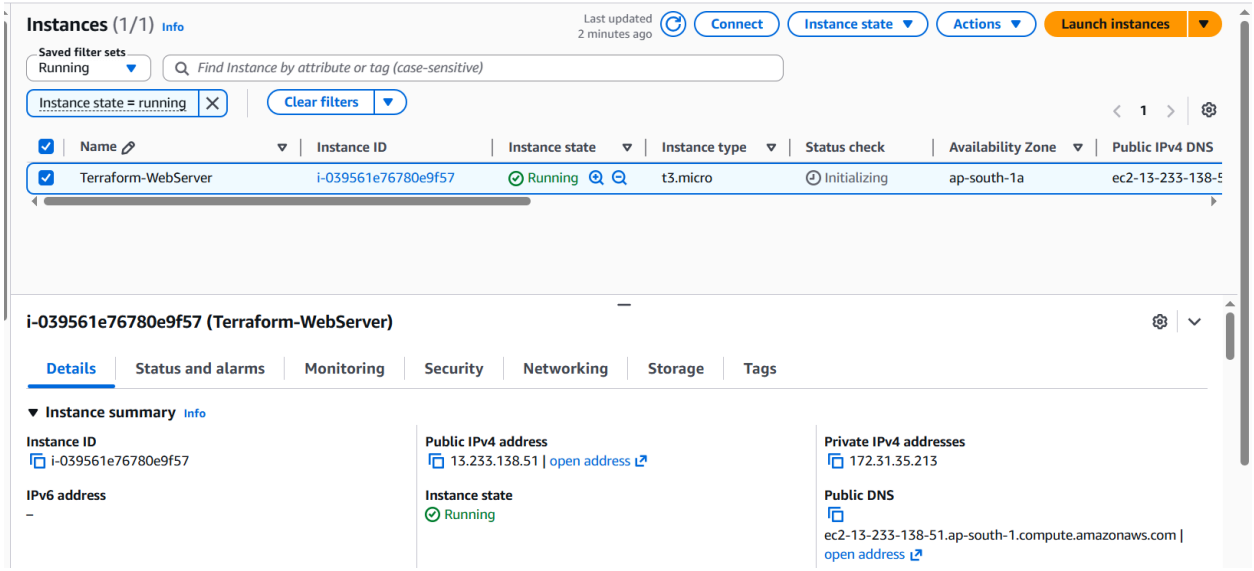
```
PS C:\Users\DELL\Desktop\Terraform> terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.web_server will be created
  + resource "aws_instance" "web_server" {
      + ami                                  = "ami-0e38835daf6b8a2b9"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
```

Ln 22, Col 2 Spaces: 4 UTF-8 CRLF { } Te

```
Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.web_server: Creating...
aws_instance.web_server: Still creating... [00m10s elapsed]
aws_instance.web_server: Creation complete after 14s [id=i-039561e76780e9f57]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Users\DELL\Desktop\Terraform>
```

Ln 22,

verify:



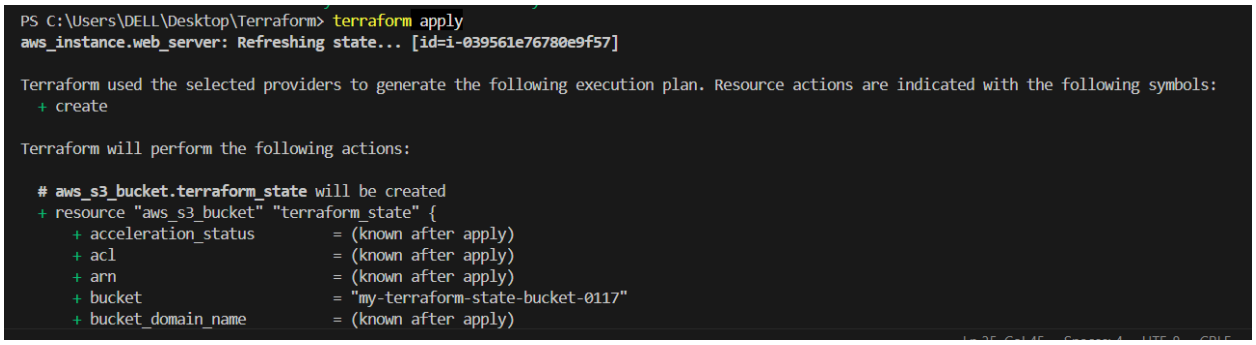
Verify in browser: public IP :80

4). Set up **S3 as backend** for task 3.

Create s3 bucket: write a script

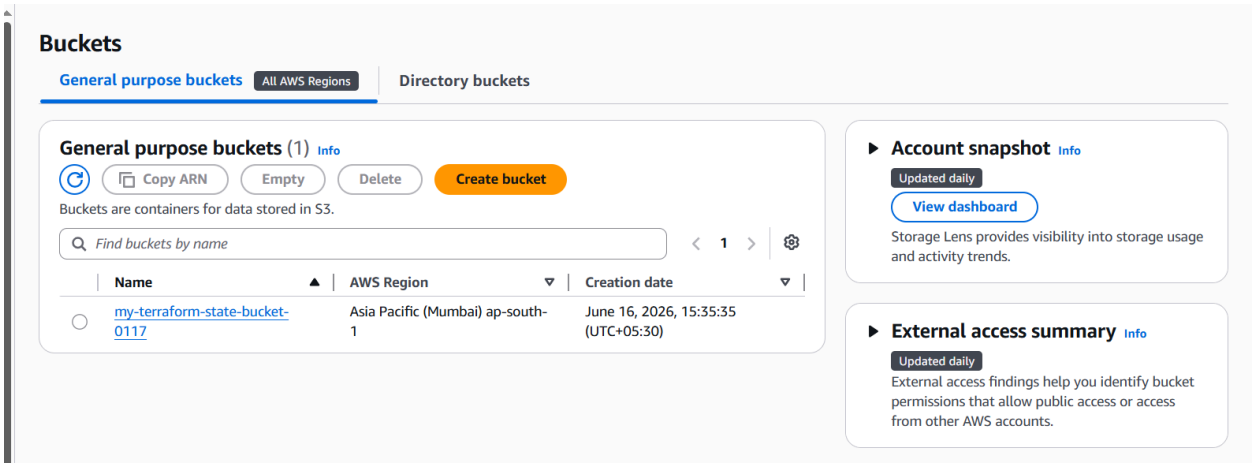
```
main.tf  terraform.tfstate
main.tf > terraform > backend "s3" > bucket
  resource "aws_instance" "web_server" {
    user_data = <<-EOF
                echo "<h1>Apache Web Server Installed using Terraform</h1>" > /var/www/html/index.html
                EOF

    tags = {
      Name = "Terraform-WebServer"
    }
  }


  resource "aws_s3_bucket" "terraform_state" {
    bucket = "my-terraform-state-bucket-0117"

    tags = {
      Name = "Terraform State Bucket"
    }
  }
```

Terraform init & terraform apply



```
PS C:\Users\DELL\Desktop\Terraform> terraform apply
aws_instance.web_server: Refreshing state... [id=i-039561e76780e9f57]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_s3_bucket.terraform_state will be created
  + resource "aws_s3_bucket" "terraform_state" {
      + acceleration_status         = (known after apply)
      + acl                         = (known after apply)
      + arn                         = (known after apply)
      + bucket                      = "my-terraform-state-bucket-0117"
      + bucket_domain_name          = (known after apply)
```

verify:



Write a script for store the state file in backend

main.tf terraform.tfstate

main.tf > terraform > backend "s3" > bucket

```
22  }
23
24
25  resource "aws_s3_bucket" "terraform_state" {
26    bucket = "my-terraform-state-bucket-0117"
27
28    tags = {
29      Name = "Terraform State Bucket"
30    }
31  }
32
33  terraform {
34    backend "s3" {
35      bucket = "my-terraform-state-bucket-0117"
36      key    = "ec2-httpd/terraform.tfstate"
37      region = "ap-south-1"
38    }
39  }
```

aws_instance Aa ab .* 1 o

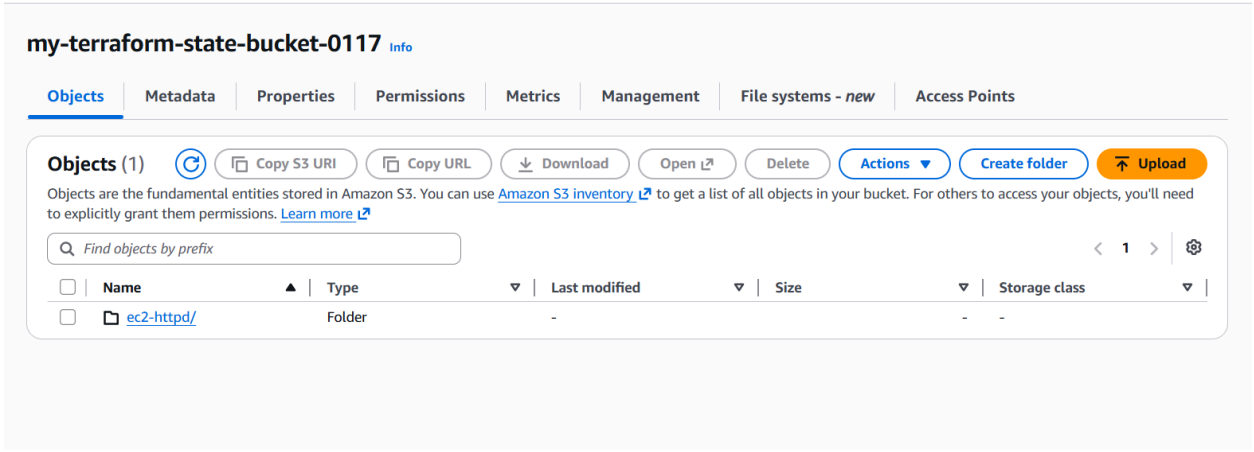
Terraform init:

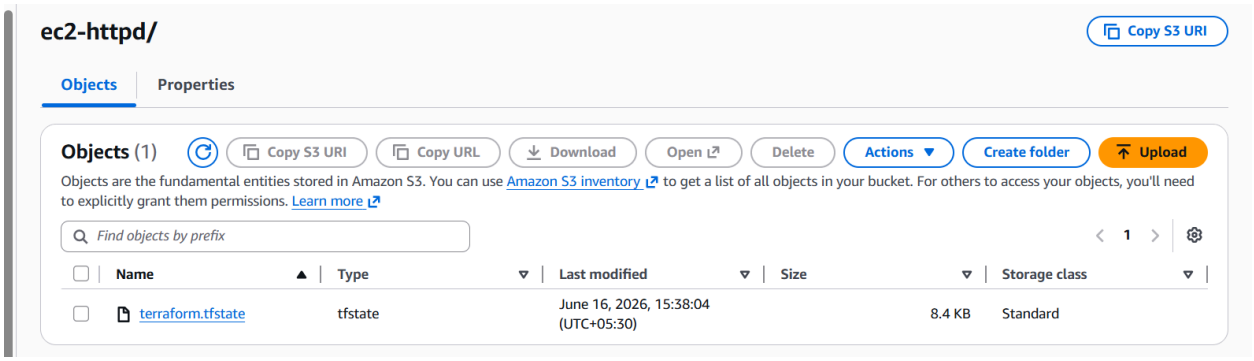
```
PS C:\Users\DELL\Desktop\Terraform> terraform init
Initializing provider plugins found in the configuration...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.50.0

Initializing the backend...
Do you want to copy existing state to the new backend?
  Pre-existing state was found while migrating the previous "local" backend to the
  newly configured "s3" backend. No existing state was found in the newly
  configured "s3" backend. Do you want to copy this state to the new "s3"
  backend? Enter "yes" to copy and "no" to start with an empty state.

  Enter a value: yes
```

Terraform apply:

```
  any changes that are required for your infrastructure. All Terraform commands
  should now work.

  If you ever set or change modules or backend configuration for Terraform,
  rerun this command to reinitialize your working directory. If you forget, other
  commands will detect it and remind you to do so if necessary.
  PS C:\Users\DELL\Desktop\Terraform> terraform apply
  aws_s3_bucket.terraform_state: Refreshing state... [id=my-terraform-state-bucket-0117]
  aws_instance.web_server: Refreshing state... [id=i-039561e76780e9f57]

  No changes. Your infrastructure matches the configuration.

  Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are n

  Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
  PS C:\Users\DELL\Desktop\Terraform>
```

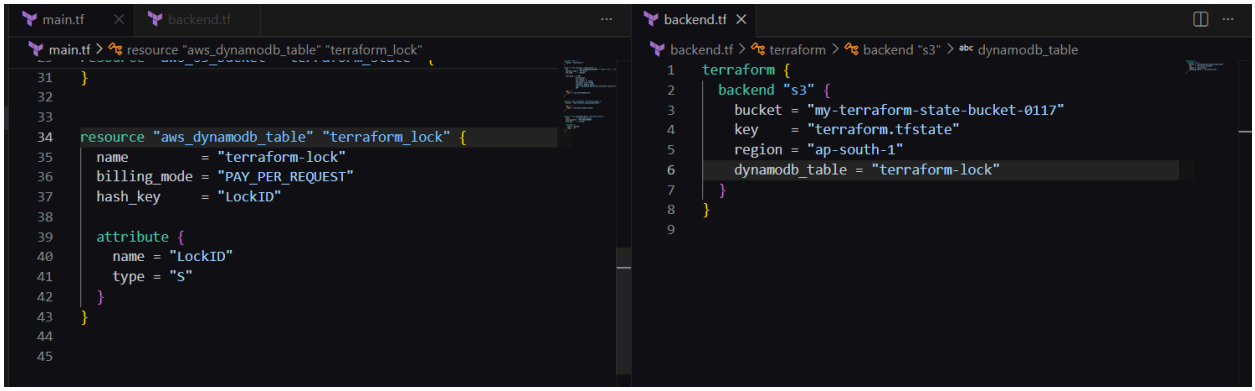
verify:



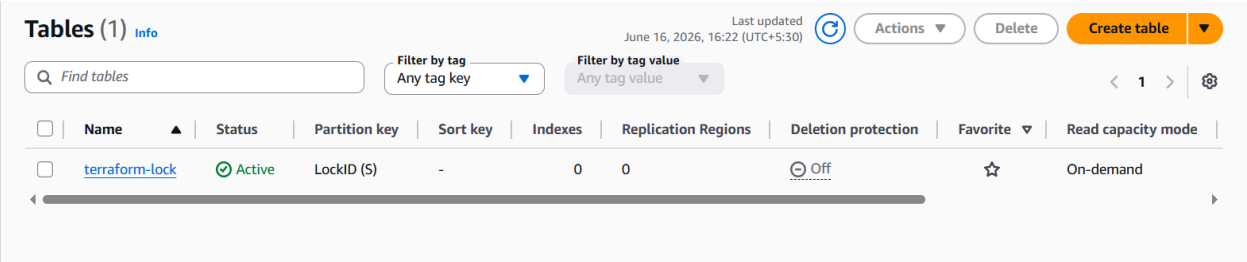


5). Set up **DynamoDB locking** for task 3.

Write a script for creating a dynamodb locking:



verify:


Tables (1) Info
Last updated
June 16, 2026, 16:22 (UTC+5:30)
Actions
Delete
Create table
Find tables
Filter by tag
Any tag key
Filter by tag value
Any tag value
1
Name
Status
Partition key
Sort key
Indexes
Replication Regions
Deletion protection
Favorite
Read capacity mode
terraform-lock
Active
LockID (S)
-
0
0
Off
On-demand